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Contained customer process
information under NDA

Alder Lake-S OEM & ODM CRD Survey Report (Desktop Segment)

Customer participated:

ACER, AS ROCK, ASUS, DELL, LENOVO, MSI, LITEON, FOXCONN, LCFC, ECS
2020 - WW45 Rev 0



Revision History

#	Rev	Date Released	Change Description
1	0	W45 Y2020	New doc

Agenda

☐ Desktop Customer Landscape

☐ Key Takeaway

☐ Additional Feedback / Concerns

☐ Observation / Next Step

☐ Printed Circuit Board (PCB) Design

☐ PCBA Factory Process

- Automation 
- Paste Printing 
- Pick and Place 
- Reflow 
- Wave Solder 
- Board Level Adhesive Process 
- ILM Cover / Lever / Assembly 
- Process Inspection / Monitoring 
- PCBA Test Process 
- PCB X-Ray 
- Rework 
- Others 

☐ Acknowledgment

☐ Back Up

2020 Desktop Customer Landscape



Q1 2020 DESKTOP OEM MSS%

Q1 '20	
Lenovo	24.4%
HP Inc	21.5%
Dell Inc	19.7%
Apple	6.9%
Acer	5.6%
Asus	5%
Others	16.8%

For Example:

- Gigabyte (1.8Mu/qtr)
- MSI (1Mu/qtr)
- AS Rock (600Ku/qtr)
- SuperMicro

Customers Providing Survey Results

OEM	ODM	Site	Design / SMT / SI	Qty/Qtr
Acer	Wistron		DESIGN	
AS Rock			DESIGN	
Asus			DESIGN	
Dell Inc			DESIGN	
Lenovo	MSI		DESIGN	
MSI			DESIGN	
Dell Inc	Foxconn	Mexico - Juarez	SI	400ku/qtr
	Foxconn	Wuhan	SMT & SI	Unknown
HP Inc	Foxconn	Wuhan	SMT & SI	600ku/qtr
	Foxconn	Czech Republic -Pardubice	SI	600ku/qtr
Lenovo	Liteon	Dongguan	SMT	400ku/qtr
	LCFC	Hefei	SMT	230ku/qtr
	MSI	Unknown	SMT	Unknown
	ECS	Shenzhen	SMT & SI	200ku/qtr

Note: For ODMs shown in blue, some limited MV/factory info was also collected

Key Takeaway (1)

□ PCB Design

- OEMs have design ownership of PCB, System and Thermal Solution.
- OSP is the most common PCB surface finish with 62 mils board thickness, followed by ~47/48mil thickness and minimum via drill diameter of 8-10mil.
- Most customer choose Type 3 board with 4/6/8 layer and 140-150°C Tg PCB lamination.
- ATX is the most common form factor design, there are also other form factors used.
- Majority of the customers have capability to support double sided assembly process.
 - ❖ Backside (secondary side) assembly process is fully automated for SMD components and capability constraints/limitations vary by customer.
 - ❖ 0201 is the smallest chip component size supported, across all board designs.
 - ❖ Component spacing rules vary by customer.
- Approximately 60% of the surveyed customers allow for trace routing between chip component pads.
- Customers have more concerns and paying more cautious for large SMT connectors, eg. include soldermask between the pins in their design to mitigate SMT risk.
- Customers perform reliability testing in board level and some in system level, the reliability tests are different among all.
- Customers use both their proprietary and Intel designs for thermal solutions. They do follow thermal relief rules.

Key Takeaway (2)

□ ODM PCBA Process & Rework

- Customer is trending to full automation for SMT assembly with in-line traceability capability set-up. Approx. 50% of the surveyed customer achieved full automation while others are partial automation.
- Pin and vacuum PCB support are used at paste print process. Approx 50% of the surveyed customer choosing pin type , while the remainder are using both pin and vacuum support.
- Customer preferred stencil thicknesses: 0.10, 0.12, 0.13 & 0.15mm for both SMT and rework.
- Pallets are generally used for board support during SMT reflow and rework. Most customers directly applied solder paste on PCB pads for rework. Few customers also print solder paste onto the solder balls.
- Approx 80% of the surveyed customers reflow with purging N2 (O2 ppm range between 1K-3K) and other reflow @ Air environment.
- No O2 ppm control for wave soldering process for most customers and PCBA rework process mostly performed at air environment.
- Many customers are not sharing their paste vendor info and machine/equipment model with Intel.
- PnP minimum ball pitch recognition capability is 0.3mm.
- **Customer prefer peak reflow temp range from 230-250°C.**
- Customers are not using adhesives for sockets but using it for BGAs. Customer common practice is to apply adhesives before board testing.



Key Takeaway (3)

□ Inspection/Test/ILM/Others

- SPI is generally performed 100% and control limits range target between 60%-180%.
- AOI is performed on 100% of components and located after reflow.
- PCBA x-ray inspections is typically performed offline by sampling, no common sampling rate. Most customers are performing bed of nails and functional tests
- Customers prefer the ILM cover to “pop off” with CPU inside socket & ILM cover closed.
- Beat rate to install or remove an Intel LGA CPU from the socket ranges between 12-15 seconds, they are concerns with socket-V lever latching at opposite direction and manufacturing run rate impact due to ILM assembly for socket-V (4-screw, 2-pcs top-plate) vs socket-H (3-screw, 1-pcs top-plate)
- Customers have not experienced any ESD issue related to the ILM cover for socket-H.
- Customers find the Desktop MAS helpful and think a System Assembly video is required. CPU assembly and PnP tool video played during Intro meeting is customers' favorite.
- Customer is looking for solution/tool that is “executable” for manufacturing process/variation (eg. ILM) and design fungibility (eg. thermal solutions)
- SJQ concerns for potential insufficient total solder volume (~10% lesser?) in solder joint formation. LOTES (0.45mm ball size) vs FIT/Deren (0.5mm ball size) but MAS recommended the same solder paste volume.

Additional Feedback/ Concerns (1)

Customer	Area of Concerns
AS Rock	Mixing & fungibility concerns, expects socket-H vs skt-V Thermal Solution is interchangeable. The hole pattern looks alike to CML-S, with distance between hole 75mm vs ADL-S is 78mm. There are additional cost and manufacturing mixing concerns when there are 2 different solutions for DT tooling, need to make it fungible due to the dimension is ~3mm delta that not noticeable by operator.
LiteON	Processor PnP tool /Stage > LiteON will be converting to automation, the PnP/assembly tool may not helpful. Automation assessment is in progress (eg. assess utilizing Cobot with tool/fixture). They plan to perform some SMT build in Oct/Nov.
Acer	SJQ concern> LOTES socket with smaller solder ball (0.45mm, 10% smaller), compare to Deren/FIT (0.5mm) but MAS recommended paste are same for FIT/Deren/OTES. Need intel to perform more data collection to ensure the same amount paste volume recommended in MAS is good for LOTES HVM. Meantime, need Intel technical explanation of why mechanical ball attached in LOTES need smaller ball & still OK to use the same paste volume & etc. Pin 1 indicator error> Different pin 1 indicator on PnP stage vs PnP tool vs PCB, ILM. Need Intel to ensure final design is correct.
Lenovo SZ (LIPC)	• Interchangeable for backplate/load plate/screw/lever & etc > Need to allow mixed vendor to ease mfg process, due to there are more pieces for socket-V (eg. allow mixed 2 vendor for lever and top plate due to 2 separate pcs). Need Intel to confirm the 4 mounting screws/nuts PN# are same. • CPU installation to socket/ILM > Need Intel to confirm any error in step, and when the cover pop-up. • "Error Free" design requirement for backplate and ILM dimension for HVM assembly> Need Intel to provide back plate dimension whether it is effective for LIPC can design an automation/fixture for backplate assembly to prevent 180° orientation (the chamfer look insignificant in picture and looks symmetrical), also potential operator installing the load plate + lever frame wrong side (vs skt-H, 3 screw, no way to install at wrong side). • Socket Pin contact dimension> Due to tighter pitch in skt-V (0.8mm), need Intel to provide pin dimension & ensure type 3 x-y axis shift always making "good" contact to the LGA pad center. • SJQ concern> Need intel to perform more validation data collection to ensure 5 mils stencil is working for LOTES HVM. • Socket & Tray> Need to ensure CTF dimension are same across all vendors, so that PnP only need to set-up 1 recipe. • Confusing Pin 1 indicator for PCB/ILM vs PnP tool/stage.

Additional Feedback/ Concerns (2)



Customer	Area of Concerns
LCFC	Run rate impact > Additional 1 screw (Skt-H 3-screw to skt-V 4-screw) and assemble 2-pcs topplate (vs skt-H 1-pcs topplate) ILM lever latch direction > Skt-H is pulling the lever inward, but Skt-V is pushing the lever outward. This is not manufacturing friendly as the same operator will be processing skt-H and skt-V. 
FXWH	• ILM > Quality control for the screw & nut thread tolerance among different vendors, variation by batch/datecode for individual vendors. • Backplate/load plate/screw/lever & etc > Need to allow mixed vendor to ease mfg process, due to there are more pieces for socket-V (eg. allow mixed 2 vendor for lever and top plate due to 2 separate pcs). • ILM gap to PCB (Warpage?) > Per skt-H experience, FXWH observed gap/separation between the ILM vs PCB. Worried if skt-V ILM will hv similar issue especially there are 2 pcs for top plate and lever pcs, if there are PCB warpage, will the situation get worsen ? And resulting poor socket contact pin to LGA pad. (eg. potential scenario with package tilted in socket, and not all pin/package unable to get the similar compressive load resulting poor or marginal contact) • Torque setting requirement and expectation > Need to “range” for manufacturing process, (eg. 8-in-lbf $\pm$ xx %) due to in actual process/production there are no way for absolute value 8-in-lbf, there are variation between batches/vendor for the screw/nuts, torque driver capability & etc.
Intel FAE	Socket bend pin pass/fail criteria in MAS > Need to re-assess the type 3 x-y axis shift visual criteria for ± 0.125 mm, due to tighter pitch in skt-V (0.8mm) vs skt-H pitch 0.914mm. Is the contact pin always making contact to the LGA pad center ? Will the contact pin that passing 0.125mm still passing electrical test ? as the pin may be touching/ too close to the adjacent CPU LGA. 

Key Message (combined both slides)

- Customer is looking for solution/tool, with design fungibility that is “error free“, “executable” for HVM process transparency or variation, (eg. opposite lever latching direction, thermal solution design, orientation control, allow mixing piece parts & etc)
- Concerns is manufacturing run rate impact for socket-V for ILM assembly (additional top plate and screw)
- SJQ concerns due to potential insufficient total solder volume for solder joint formation
 - LOTES (0.45mm solder ball) vs FIT/Deren (0.5mm solder ball) but MAS recommended same solder paste volume



Observation & Next Step

□ Observation

- Few customers were not excited about Intro meeting (eg. ECS, MSI, HPI, QCMC) or participate in CRD survey (eg. QCMC, Gigabyte, MSFT)
 - ❖ Customers already get used with the working model with Field Engineer/account team for ~10 years without CQR/CME enabling

□ Next Step

- Demonstrate our values and collaborate with Field Engineer/account team for future customer enabling in DT segment
- Proactively follow-up with customer for their development activities/NPI for Socket-V and offer help to re-build customer relation
- Follow-up with LiteON for their Oct/Nov SMT build result and automation assessment
- To seek clarification from customers for items label in **this color**



Acknowledgment

Customer	Account CQE
Acer	Jeremy Chang
AS Rock	Sean Bao
Asus	David Wang
Dell Inc	Allan Leal, Samer Alaiyoubi, Arturo Rios
ECS	Sean Bao
Foxconn WH	Sean Bao
HP Inc	John Burtchaell
LCFC	Shirley Zhang
Lenovo	Jiang Art, Albert Tsai
Liteon	Sean Bao
MSI	David Wang, William Pan
QCMC	Juanita Ren

Back Up

Printed Circuit Board Design (1)

Area of Interest	LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Do you have PCB design ownership	Yes	Yes	Yes	Yes	Yes	Yes
Do you have Full System Design Ownership	Yes	Yes	Yes	Yes	No, only for channel MB	No, Motherboard only
Do you have Thermal Solution Design ownership	Yes	Yes	Yes, for AIO/miniPC, No, for Tower (Owned by Acer)	Yes	Yes	Yes, but only for PCH and Vore MOS heatsink
Please list the PCB suppliers used by %	GBM, APCB, Hanstar, Junya (駿亞), ECS (15%), Palwonn, VGA, etc...	Not Answered	4 Layer: CEE(1st) / GLOBALBRAN(2nd) / VICTORY(3rd) 6 Layer: Hannstar(1st) / CEE(2nd) / GLOBALBRAN(3rd) 8 Layer: Hannstar(1st) / GLOBALBRAN(2nd)	TRUSTECH(40%), GECS(30%), DF(15%), DYEC(15%)	Vendor A(30%), VendorB(70%)	Trustech(50%), Dingfu鼎富(50%)
Please list the PCB surface finishes used by %	OSP	OSP(100%)	OSP(100%)	OSP(100%)	OSP(100%)	OSP(100%)
Please list the PCB board thicknesses used by %	62mil(90%), 48mil(10%)	62mil(70%), 47mil (30%) All of the "Micro" form factors in Dell's Desktop portfolio use 47 mil (1.2mm) thick PCBs. This equates to ~4-5 million units of volume and ~30% of their Desktop volume. All other Dell Desktop products use 62 mil (1.6mm) thick PCBs, which equates to ~70% of their Desktop volume.	62mil(75%), 47mil(25%)	62mil(100%)	62mil(80%), 59mil(20%)	62 mil(90%), 59 mil(10%)

Key Message

- OEMs have design ownership of PCB, System and Thermal Solution
- OSP is the most common PCB surface finish
- While 62mil is the most common PCB thickness, ~47/48mil are also commonly used by 3 customers

Printed Circuit Board Design (2)

Area of Interest	LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Please list the PCB Laminate Tg used by %	150°C Tg(50%), 140°C Tg(50%)	150°C Tg(100%)	145°C Tg(100%)	150°C Tg(50%), 140°C Tg(30%), 170°C Tg(20%)	150°C Tg(100%)	140°C Tg(80%), 150°C Tg(20%)
Please list the PCB type that is used by %?	Type 3(100%)	Type 3(100%)	Type 3(100%)	Type 3(100%)	Type 3(100%)	Currently use Type 3(100%), but we will validate type 4 for PCIe5
Please list the PCB layer count that is used by %	4 Layer(65%), 6 Layer(30%), 8 Layer(5%)	4 Layer(60%), 6 Layer(25%), 8 Layer(15%)	6 layers(50%), 4 layers(25%), 8 layers(25%)	6 Layer(40%), 4 Layer(30%), 8 Layer(30%)	6 Layers(70%), 4 Layers(20%), 8 Layer(10%)	Currently use 6 Layer(100%), but we will validate 4 Layer PCB
Please list the PCB board form factor that is used (by %)?	Non-standard sizes, e.g. 267mm x 245mm, 267mm x 220mm, 170mm x 170mm, etc...	Custom(100%)	mATX(25%), DTX(25%), AIO(25%), miniPC(25%)	ATX(80%), uATX(20%)	ATX(43%), MATX(33%), ITX(19%), EATX(5%)	Our first project is ATX. Usually we might have ATX(40%), uATX(30%), mini-ITX(15%), and AIO(15%) in the future.
What % of your designs are double sided assembly ?	Double sided(90%), Single sided(10%)	Double sided(90%), Single sided(10%)	Single Sided(90%), Double sided(10%)	Double Sided(80%), Single Sided(20%)	Double Sided(100%), Single Sided(0%)	Single Sided(70%), Double Sided(30%),
Is there capability to support 100% of your volume with double sided components?	Should be OK	Yes	Yes	Yes	Yes	Yes, we can support 100% double sided components

Key Message

- 140-150°C Tg PCB laminate are the most common
- Type 3 and 4/6/8 layer PCB designs are the most common
- While ATX is the most common form factor design, many other various form factors are also used
- It is most common for customers to have the capability to support double sided assembly

Printed Circuit Board Design (3)

Area of Interest		LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Via & Pad	What is the minimum via pad stack (drill diameter / via pad diameter / antipad diameter)?	10mil drill / 20mil pad / 31mil antipad	8~10mil drill / 18mil pad / 28mil antipad	8mil drill (Finish hole size) / 18mil pad / 28mil antipad	10mil drill / 20mil pad / 30mil antipad	12mil drill / 22mil pad / 32mil antipad	10mil drill / 18mil pad / 28mil antipad
	Are the BGA land area vias not covered with soldermask, or are they capped / plugged / covered with soldermask (no exposed vias)?	covered with soldermask	both are used (covered & not covered with soldermask)	covered with soldermask	covered with soldermask	plugged / covered with soldermask	Full covered with soldermask (全塞孔)
Component Sizes	What is the smallest chip component size that is supported for your desktop board designs?	some are 0201 and some are 0402	0201	0201	0201	0201	0201 is OK
	Regarding the above question, is this true across all board designs (for example, across all 4 or 6 layer board designs, and across all Type 3 and Type 4 designs)?	Yes, across all board designs	Yes, across all board designs	Yes, across all board designs	Yes, across all board designs	Yes, across all board designs	Yes, across all board designs

Key Message

- 8-10mil is the most common minimum via drill diameter
- 0201 is the smallest chip component size supported, across all board designs

Printed Circuit Board Design (4)

Area of Interest		LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Component Placements	Is backside assembly a fully automated process	Fully automated process	Fully automated process	Fully automated process	Fully automated process	Fully automated process	Fully automated process
	Any limitations on backside SMT component placement (quantity, chip size)?	<u>Part Height:</u> Needs to be less than 3.5mm. <u>Constraints:</u> Cannot put connector / socket for most product, need to be far away from the PCB edge (more than 6mm).	<u>Quantity:</u> Balance component quantity for top and bottom side	<u>Chip Min:</u> 0201 <u>Component Max size:</u> 100mm*90mm <u>BGA:</u> <u>Max pitch:</u> 0.5mm <u>Size:</u> 5x5mm to 45x45mm <u>Smallest Diameter:</u> 0.25mm <u>QFP:</u> <u>Size:</u> 5x5mm to 45x45mm <u>Smallest Spacing:</u> 0.4mm <u>Smallest Lead Width:</u> 0.2mm	<u>Part Height:</u> Only the part height is limited to 6mm on backside <u>Quantity:</u> The quantity is not limited	<u>Placement:</u> Backside SMT components must be placed away from the edge of board by at least 200mil (5mm) <u>Via/Through Hole:</u> Distance between "VIA & Through hole (PTH & NPTH)" and SMD PAD must be equal or greater than 6mil (0.15mm) <u>Vias:</u> It is strictly prohibited to lay VIA hole (including PTH) beneath SMT component <u>Part Height:</u> -Backside component height must be equal or less than 12mm for double-sided process board -If SMT component height (t) ≤ 2 mm, distance between backside DIP pin and SMD component (x) must ≥ 3 mm. -If SMT component height (t) ≥ 2 mm (The height can't over 12mm), distance between backside DIP pin and SMD component (x) must ≥ 4 mm. <u>Quantity:</u> And the component Q'ty should be ≤ 1000 pcs	<u>Quantity:</u> No rule, so far the maximum backside component quantity is 1243 <u>Chip size:</u> No rule, so far the maximum backside component is Intel LGA1200 socket <u>Beat rate:</u> The backside SMT will use a slower pick & place machine.

Key Message

- The backside assembly process is fully automated for all customers
- Backside component constraints & limitations vary by customer

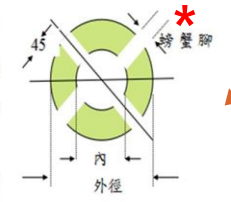
Printed Circuit Board Design (5)

Area of Interest		LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Component Spacing	Min spacing for 0402 to 0402	Min spacing: 1.5mm, 2mm is better	pad-pad ≥0.3mm (12mils)	0.25mm	0.3mm (12mils)	0.05mm	0.15mm
	Min spacing for 0603 to 0603	1.5mm	pad-pad≥0.3mm (12mils)	0.30mm	0.3mm (12mils)	0.05mm	0.18mm
	Min spacing for 0805 to 0805	1.5mm	pad-pad≥0.3mm (12mils)	0.40mm	0.3mm (12mils)	0.05mm	0.18mm
	For through-hole components (e.g. leaded caps), what is the minimum body-to-body spacing requirements for assembly?	≥0.5mm (20mil)	2.54mm to 3.0mm (100 to 120mils), ODM depedant	2mm	minimum body-to-body spacing requirements is 0.61mm (24 mil)	0.50mm each cap frame spacing (≥0.5mm (20mil))	0mm between component allegro symbol outline
	For large SMT components (e.g. inductors), what is the minimum body-to-body spacing requirements for assembly?	≥0.5mm (20mil)	1.0mm to 2.54mm (40 to 100mils), ODM dependant	2mm	minimum body-to-body spacing requirements is 0.30mm (12 mil)	Not answered	0mm between component allegro symbol outline
	For components inside the socket cavity, what is the minimum chip-to-socket cavity-wall spacing requirements for assembly?	≥0.5mm (20mil)	2mm	1mm	minimum chip-to-socket cavity-wall spacing is 1.0mm (40 mil)	All the caps around the CPU SKT should be >0.2mm	0mm between component allegro symbol outline
	For the above question, is this minimum spacing set for rework reasons or for other assembly considerations? (please explain)	Above spacing is set for request for rework limitation and progress assembly in MFG	Above spacing is set for both rework and assembly considerations	Above spacing is set for Chip to BGA 2mm	Above spacing is set for rework reasons	Above spacing is set for: Component Placement, Assembly, Installing, Crashworthiness and maintenance considerations	Above spacing is set for factory assembly

Key Message

- Component spacing rules vary by customer

Printed Circuit Board Design (6)

Area of Interest		LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Thermal Reliefs	Do you follow any rules / requirements for thermal reliefs?	Yes	Yes	Not Answered	Yes, we follow Intel's KOZ Limit, and by model (Mass / Height, etc...)	Yes	Please refer to the graphic below the materials are to be shipped Thermal Relief 尺寸設計規範: 單位為 mil ✓ 其 Flash Name 為 TR_aXbXc-d a 表示內徑 : DRILL-SIZE + 16 b 表示外徑 : DRILL-SIZE + 30 c 鍍錫腳距 : 12(當 DRILL-SIZE = 10 以下) 15(當 DRILL-SIZE = 11 ~ 40) 20(當 DRILL-SIZE = 41 ~ 70) 30(當 DRILL-SIZE = 71 ~ 170) 40(當 DRILL-SIZE = 171 以上) d 鍍錫腳角度 : 0 or 45 *TR 0X0X0-0 指的是 FULL CONTACT(全導通)  *鍍錫腳角度 : 0 度 ✓ 其 Flash Name 為 TR_aXbXc-0 適用零件 : 1. 針對 Connector-2 排以上交錯式排列。 2. 針對 Chipset-PGA 類型。  功能: Thermal Gap Angle 為改為 0 度, 其導通性較佳。
Trace Routing for Passive Components	Do you allow for trace routing between the pads, or underneath the pads, of chip components, such as 0603 and 0805, etc...?	No, not suggested, because of the layout solution, due to interference problems. If were to modify the trace width to through the pad, the impedance may not meet the SPEC.	Yes	No, concerns: 1. The signal performance will be poor. 2. It will cause a short issue due to poor soldermask coverage 3. It will cause poor solderability due to trace thickness is higher than the pin	Yes	No, must consider soldermask application, especially in the dark soldermask / registration	Yes, but only when there is not other solution

Key Message

- Customers do follow thermal relief rules
- 3 customers allow and 2 customers don't allow for trace routing between chip component pads

Printed Circuit Board Design (7)

Area of Interest		LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Trace Routing for Large SMT Connectors	We see a trend with higher speed IO to move to large SMT connectors (e.g. PCIe, DDR UDIMMs). Do you see any issues or concerns with these types of connectors?	Yes, reliability is a concern for a long slot, such as PCIe Gen5 and DIMM DDR5. Another concern is that it will add more VIAs and layers.	Need to evaluate case by case	There are no concerns for Desktop, this concern should be for Server. Basically, we will follow Intel's Demo board design. Meanwhile, we will check with our EE and ODM.	Need extra VIA for routing trace to TOP layer	Yes, wait for test	Yes. 1) The factory assembly failure rate is much higher than DIP type slot, and 2) SMD type is difficult to rework / repair.
	For large SMT connectors (e.g. PCIe, DDR UDIMMS, etc...) with pitch ~0.85mm to 1.0mm, do you allow for soldermask between the pins or are ganged soldermask openings a requirement? (see pic)	We are forbidden to do this type of design.	Need to have 0.20mm (8 mils) air gap from trace to adjacent pin pad	Trace between pins must be covered with soldermask to prevent short issue	Need to allow for soldermask between the pins	Need to cover between the pins with soldermask	Alder Lake design is still in Layout process, but we think soldermask is a must for production quality control. We can route between the pins with minimum 1. Trace width = 0.0889mm (3.5mil) 2. Pad width = 0.1778mm (7mil) 3. Pad to trace = 0.12065mm (4.75mil)
	For the above question, Intel is planning to route traces between these pins (please refer to graphic on the right), do you have any concerns?	We are forbidden to do this type of design.	Soldermask is required to prevent short during assembly process	The spacing between the pin and trace must be larger than 0.1143mm (4.5mil), and the minimum trace width must be at least 0.0889mm (3.5mil) at least	Yes, It's more challenging to routing the DDR5 trace for 4 Layer PCB designs	Pin to line gap must be 0.127mm (5mil) minimum to avoid shorting risk	For 2DPC DDR5 design, Intel PDG still shows TBD and no RVP board file available for us to download. We will need to try this routing by ourselves. From Intel's documentation: "DDR5 UDIMM 2DPC Design Guidelines are TBD"

Key Message

- Customers tend to be cautious and have some concerns with large SMT connectors
- If large SMT connector pins are used, Customers need to have soldermask present between the pins

Printed Circuit Board Design (8)

Area of Interest		LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Reliability	Do you perform any reliability validation on your sockets?	Yes	Yes	Yes	Yes	Yes	Yes
	Are your reliability tests performed at a board and/or system level?	Both Board & System Level	Both Board & System Level (For Board level we use daisy chain test vehicles)	Both Board & System Level	Board Level	Board Level	Board Level
	Do you use the Intel reference thermal solution (heatsink design) or proprietary design?	Proprietary design for thermal solution	Proprietary design for thermal solution. It depends on the amount of heat dissipation required vs fan speed to remove the heat. Still being reviewed.	Acer is using both the Intel reference thermal solution (for mobile) and proprietary design (for special form factors). The Tower solution is Acer's AVAP (AVAP is Acer's assigned Vendor).	Intel design for thermal solution	Proprietary design for thermal solution	Proprietary Design for VR Not Answered for thermal solution
	If you use proprietary thermal solutions could you provide any further information on your thermal solution characteristics?	Thermal solution characteristics are case by case	Thermal solution characteristics are form factor dependent. Thermal Engineering still reviewing options. Dell doesn't have a minimum thermal solution range and they typically don't exceed 550g. But for ADL-S (125W SKU) Dell is looking at going up to a ~900g thermal solution.	AIO (All in One) and BOX (small form factor like Intel's NUC design) will follow Intel's rules and design requirements for Thermal solution characteristics. If no specific design, will follow NB standard design for Thermal solution.	By model (Mass/Height etc.) and Center of Gravity (CG), will follow Intel's design for thermal solution	Thermal solution characteristics: aluminum design, extended heatsink, thermal pads.	Not available yet.

Key Message

- Customers perform board level reliability testing (some also perform system level testing)
- Customers use both proprietary and Intel designs for their thermal solutions

Printed Circuit Board Design (9)



Area of Interest	LENOVO (MSI)	DELL	ACER (WISTRON)	ASUSTEK	MSI	ASROCK
Reliability	Which type of reliability tests and stress levels do you perform and expect to meet? Please list and describe each test. 1. Guardband: 0°C, 25°C, 60°C Burnin for 20 mins 2. IC Guardband test: 0°C, 25°C, 60°C Burnin for 1 hour 3. Storage temperature & humidity test: -40 ~ 60 °C for 96 hours 4. System Operation Vibration Test: Grms = 0.27, 30 minutes per axes. 5. System Fragility Vibration Test: Grms = 1.04, 15 minutes per axes. 6. System Operation Shock Test: 3ms(15G) for 4 Axis (+X, -X, +Y, -Y), 3ms(30G) for 2 Axis(+Z, -Z), half-sine wave, each side will do 1 times. 7. System Fragility Shock Test: 45G/11ms, Trapezoidal wave, 6 Sides, each side shock one time. 8. High/Low temperature: Run 3Dmark 8hour under 40°C / 0°C 9. Etc...	<u>Board level:</u> 1. HALT Test: Temp (-15C to 75C) with 6 minutes Dwell, Vibration (30Grms, 5min). Measuring all chains contact resistances for continuity 10 cycles. Pass criteria: no opens 2. Temperature/Voltage Margin Test 3. Hot & Cold Operation (40C 24hrs, 0C 24hrs) 4. High Humidity Operation (32C 80% 24hrs) 5. Power Cycling Test (40C 80% 1650 cycles, 0C 1650 cycles) 6. Thermal Cycling: Temp (0C to 100C), 1000 cycles. Measuring all chains contact resistance for continuity. Pass criteria: no opens. <u>System level:</u> 1. Non-Operational Random Vibration (1.37Grms, 15min) 2. Non-Op Half Sine Shock Test (105G,2ms) 3. Strain Measurement Test SetUp (105G,2ms) 4. Hot & Cold Operation (40C 24hrs, 0C 24hrs) 5. High Humidity Operation (32C 80% 24hrs) 6. Power Cycling Test (40C 80% 1650 cycles, 0C 1650 cycles)	Both reliability tests and stress levels Acer will provide additional details, and Intel can then provide the feedback (Mak to follow up w/ Morris Yang).	1. Non-Operation Shock Test: Trapezoidal Shock (Square Wave Shock) a) Maximum faired acceleration: 50G b) Velocity change: 170 inches/sec c) Test direction: 6 orientation face 2. Non-Operation Vibration Test: a) Non Operating Random Mode b) Fixture used (wooden or plastic): fasten the unit to the table. c) 10 min/axis Vibration Frequency Levels Tested: 1) 5 Hz (Frequency), "- dB/Oct (Slope), 0.01 G squared/Hz (PSD) 2) 20 to 500 Hz (Frequency), "- dB/Oct (Slope), 0.02 G squared/Hz (PSD) 3. Thermal Shock Test - Non-OP: -40°C~85°C, 1.5hr/cycle, 27 cycles	1. Windows Shutdown 2. Xcopy 3. Heaven Benchmark 4. 3DMark 5. Windows reboot 6. S3 7. Shock 10 times 8. Vibration (X, Y, Z) 0.5 hrs 9. etc...	1. Shock 2. Vibration

Key Message

- Customers perform vary different reliability tests

PCBA Factory Automation



Process	ECS	LCFC	FXWH	LITEON
PCBA Automation	Yes	Yes	Yes	Yes
Full or Partial automation	Full	Partial	Full	Partial
Test Automation (in-line or off-line)	In-line	In-line	In-line	In-line
Automation machine	Panasonic SMT machine TRI SPI/AOI	mounter, printer, SPI, reflow, AOI, ATE	not answered	SMT: NXT Gen I, II, III Panasonic :NMP&CM602 DIP:CY-450(超越)
What is used to pick Intel's component	vacuum nozzle	vacuum nozzle	vacuum nozzle	GPRO SMT -- Intel PCH / LAN Chip, vacuum nozzle
Purpose of scanning 2D Matrix	traceability	traceability, inventory check	traceability	traceability, inventory check
Which process require 2D scanning ?	IQC/SMT	SMT	incoming inspection to outgoing shipment	IQC, SMT, DIP, FPT
In-line or Off-line traceability?	In-line	In-line	In-line and Off-line traceability	Both In-line & Off-line

Key Message

- For PCBA factory, 2 Customers have full automation and 2 Customers have partial automation
- In-line traceability capabilities are most common to have across the customers

Paste Printing Process



Parameter / Capability	ECS	LCFC	FXWH	LITEON	MSI (LENOVO)
Squeegee Speed (mm/sec)	60-120	120	80-90	100	50
Squeegee Pressure (Kg/area?)	4.5-8kg/cm2	2	8~10g/mm	10	10
Stencil Separation Speed (mm/sec)	0-3	1	1	2	not answered
Under Screen Cleaning frequency (per print)	2-6	5	1	6	not answered
Stencil Cleaning Frequency (hrs)	1	4	6	100	not answered
Cleaning Process	Pneumatic	Solvent clean	Ultrasonic	Manual wiped paper & Ultrasonic	not answered
PCB support type	Pin, Vacuum	Pin	Pin, Vacuum	Pin, Vacuum	Pin
Do you use the same pallet for paste print, PnP and reflow?	no	no	no	yes	not answered
Maximum board handling (mm, length x width)	510 x 508.5	460 x 360	yes	610 x 510	not answered
Stencil Cut / Coating	Laser	Nano coating	yes	Electroform, Laser cut with electro polish + nano coat	Etching Laser. No electropolish.
Stencil Thickness (mm)	0.13	0.13	0.10, 0.13, 0.15	0.10 (BGA), 0.13 (skt), 0.15 (other)	Secondary: 0.12/0.15 Primary: 0.12
Stencil Step up/down (mm)	Yes 0.10/0.15	no	Yes. Depends components, 0.10, 0.13, 0.15	Yes. Depends components, 0.10 (BGA), 0.13 (skt), 0.15 (other)	not answered
Solder Paste Vendor	not answered	not answered	Eunow EUP-148	not answered	Shenmao PF606-P
Solder Paste Type	type 4	SAC type 4	SAC305 type 4	Type 4	SAC305 Type 4

Key Message

- Pin PCB support type is most common and several customers also using vacuum support
- Customer preferred stencil thicknesses: 0.10, 0.12, 0.13 & 0.15mm
- Many customers are not sharing their paste vendor info with Intel

Pick & Place Process



Parameter / Capability	ECS	LCFC	FXWH	LITEON	MSI (LENOVO)
Nozzle size (LGA socket & Intel component)	1005	NPM 1003	Universal 1240F & Fuzion1-11 and Sony CF00900	Nozzle: 15.0G (skt) Nozzle: 7.0G (BGA)	FUJI NXT1. Nozzle types 0.7/1.0/1.3/2.5/5.0/7.0 /10.0
Max Socket & BGA size capability (x,y, mm)	90-120	45 x 45	55 x 55	74 x 74	not answered
Max Socket & BGA weight capability (grams)	28mm ?	30	not answered	60	not answered
What component placement force is used for Intel components?	0.5mm ?	0.3mm?	150g	N/A	Factory target is 0
What method of board support is used during placement?	Tray	Pin	Pin	Pin	not answered
Maximum board handling capability (length x width, mm)	510 x 590	Dual land : 510 x 300 Single land : 510 x 590	610 x 813 (universal)	534 x 510	not answered
Min pitch/ball recognition capability (mm)	0.3-1.5	Not answered	0.4	0.3	not answered

Key Message

- Minimum ball pitch recognition capability is 0.3mm

Reflow Process (1)

Parameter / Capability	ECS	LCFC	FXWH	LITEON	MSI (LENOVO)
Yield target	99.9% (1K DPM)	Not Answered	99.9% (1K DPM)	98%	not answered
Do you use a PCB support pallet at reflow?	Yes	No	Yes & No	Depends (see below)	No
Is the PCB support pallet decision based on PCB thickness or something else?	No	N/A	No	Yes - Pallet used for thin/big PCB	No
PCB support pallet material?	Synthetic stone	N/A	Not Answered	Synthetic stone (合成石)	not answered
Are PCB support pallet tooling pins used, or can the board move in the X or Y direction?	No pins used	N/A	Yes, pin support	Yes, pin support	not answered
PCB support pallet minimum board clearance to pallet edge? (mm)	Not Answered	N/A	No ?	1	not answered
PCB support pallet, type of clamps used?	Not Answered	N/A	No clamps used	Not Answered	No clamps used
Oven Purging with N2?	Yes	Yes	No	Yes	Yes
O2 PPM control	<1000ppm	<3000ppm	N/A	<3000ppm	<1000ppm

Key Message

- 4 Customers using N2 (1K-3Kppm O2) @ Reflow and 1 Customer using Air @ Reflow

Reflow Process (2)



Parameter / Capability	ECS	LCFC	FXWH	LITEON	MSI (LENOVO)
Belt Speed (cm/min)	85	130	85-90	100	105
Ramp Rate (°C/sec)	1-3	1-3	1-3	1-3	not answered
Cooling Rate (°C/sec)	-1 to -3	-1 to -3	-1 to -3	-2	not answered
Peak Temp Range (°C)	240-245	230-250	235-250	235-245	235-250
Regarding the question above, if you must run above 245°C, please explain the reason.	N/A	1. oven character 2. reflow profile needed	N/A	N/A	not answered
Time above Liquidus (sec)	72-87	60-90	60-90	60-90	60-100
Soak Temp Range and Time (sec)	85-105	60-100	60-120	60-90	60-120
How many TC and what loc's or used on the Intel Parts during profiling?	8 (BGA, IC, Chip, PCB)	10 (CPU, GPU, RAM)	3	5	Follow Intel's recommendations
What is the delta T across the Intel component/socket area? (°C)	<10	not answered	<10	<10	not answered
What is the delta T across the whole motherboard? (°C)	<15	<10	<15	<15	not answered
What is the number of heating and cooling zones used?	12 heating, 2 cooling zones	13 heating, 4 cooling zone	Heller 1913. 12 heating, 2 cooling	13 heating, 4 cooling zones ?	Heller 1900EXL. 12 zone oven
Maximum board handling capability (length x width, mm)	457 x 510	width 500	406 x 610	width < 460	not answered

Key Message

- Customer's reflow belt speeds range from 85 to 130cm/min
- Customer's peak reflow temp range from 230-250°C
- Customer's delta T across the Intel component/socket area <10°C



Wave Solder

Parameter / Capability	ECS	LCFC	FXWH	LITEON
Preheat Time (sec)	60-120	120	60 - 80	90-150
Solder Pot Temperature (°C)	260 - 270	260 - 270	270 – 310	260 - 270
Conveyor Speed (cm/min)	100.8	80	120 – 160	100 -140
What are the components and pitch that undergo wave soldering?	DIP plug-in	not answered	not answered	DIP Parts is OK.
Are components hand placed or machine placed?	Manual & Auto – component dependent	Manual	Manual	Manual & Auto – component dependent
What is the O2 PPM control?	no control	no control	no control	no control
Yield Target	99.9% (1K DPM)	not answered	99.9% (1K DPM)	98%

Key Message

- Customer's wave conveyor speeds range from 80-160 cm/min
- There is no O2 ppm control for wave

Board Level Adhesives



Process / Capability	ECS	LCFC	FXWH	LITEON
Are you currently using adhesive for Sockets?	No	No	No	No
Are you currently using adhesive for BGAs?	Yes	Yes	Yes	Yes. Depends on different BGA.
What are the primary reasons adhesive is applied?	strengthening	solder crack	N/A?	N/A?
What type of adhesive do you use?	Corner glue	Under fill ?	Corner glue	Corner glue?
Is the adhesive dispensed manually (by operator/hand) or automated (by machine)?	Automated	Automated	Automated & Manual	Automated & Manual. Depends on the production forecast.
What is the vendor name and model of adhesive material you are using?	Easyseal (易品) E-8369	AXXON	ME	EAST PIVOT(易品) and MODEL:4058-2 Series
Is the adhesive reworkable?	Not answered	Yes	No	Yes
Where in the board manufacturing process is the adhesive applied?	After SMT/Before Test	Before Test	After SMT/Before Test	Before DIP (wave)

Key Message

- Customers are not using adhesives for sockets, but they are using it for BGAs
- It is most common for customers to apply adhesives before Test

ILM Cover/Lever/Assembly



Parameter / Capability	ECS	LCFC	FXWH	LITEON
Do you have any feedback on the Socket V ILM Cover design features or markings?	Yes. Logo?	No	not answered	not answered
What is your ergo spec (maximum limit) for install or remove ILM Cover (all 4 snap features) by manual operator?	by manual operator	not answered	not answered	Torque 5KG via electrical screw driver
Do you prefer the ILM Covers to be recyclable (per ISO 11469)?	No	not answered	Yes	No preference
Should the ILM Cover "pop off" (disengage) when an Intel CPU is placed inside the socket and the ILM cover is closed?	Yes	not answered	Yes	Yes
Did you see any ESD failures related to Socket H ILM Cover?	No	not answered	No	No
Do you use a fixture for the assembly of the ILM components? If yes, what kind of fixture?	Yes, fixture	not answered	No (why no?)	Yes, fixture
What is your ergo spec (maximum limit) for opening or closing the ILM lever by manual operator?	not answered	not answered	not answered	Torque 5KG via electrical screw driver
What is the time required for attaching the ILM Assembly onto the motherboard? (sec)	24	not answered	not answered	18

Key Message

- Mixed feedback regarding ILM cover recyclability requirement
- Customers prefer the ILM cover to "pop off" with CPU inside socket & ILM cover closed
- Customers have not experienced any ESD failures related to the socket H ILM cover

Process Inspection / Monitoring



Process Parameter / Capability		ECS	LCFC	FXWH	LITEON	MSI (LENOVO)
SPI (Solder Paste Inspection)	Sampling rate	100%	100%	100%	2 PCB/2.5hrs	not answered
	Machine	not answered	not answered	Holly S8030	not answered	MSI-AOI
	Min/Max Spec and control limit	60%-180%	not answered	50%-180% (paste area). 50%-160% (paste height).	0-0.3mm (0,13 skt, 0.1 BGA) ?	60%-150% (paste area). 60%-180% (paste height).
	Is SPI Performed on selective components or 100%?	100%	100%	100%	Selective	not answered
	What is the typical percentage rate of board wash?	0.10%	not answered	not answered	not answered	not answered
AOI (Automated Optical Inspection)	Maximum board handling capability (length x width, mm)	510 x 460	width 500	510 x 460	460 x 510	not answered
	In-line / offline ?	In-line	In-line	In-line	In-line	not answered
	Sampling rate ?	100%	100%	100%	100%	not answered
	How many AOI's are used?	not answered	2	2	1pcs/ 1 line	not answered
	Where is AOI located?	After reflow	after reflow/ before wave	After reflow	After reflow	not answered
	Is AOI Performed on selective components or 100%?	100%	100%	100%	100%	not answered
	Maximum board handling capability (length x width)	510 x 460	width 500	510 x 460	460 x 510	not answered

Key Message

- SPI control limits range between 60%-180%
- AOI is performed on 100% of components and located after reflow

PCBA Test Process



Process Parameter / Capability		ECS	LCFC	FXWH	LITEON
Bed of Nails Test	For test, are you doing any kind of bed of nail testing?	Yes	Not answered	ICT	Yes
	For the above question, on which side of board is the testing being done?	Bottom & Top	Not answered	Bottom	Bottom & Top
	How many Bed of Nails tests are used?	Maximum support 3584 test nails	Not answered	1800 - 3000	Depend on production
	Where is the Bed of nails test located?	After dip before functional test	Not answered	ICT	After re-flow. Before function test.
	Is Bed of Nails test Performed on selective components or 100%?	not 100% , need detail mode to analysis	Not answered	1	Required 90%. In actual depend on production
	Maximum board handling capability (length x width, mm)	520 x 400	Not answered	388 X 320 (Matira5)	500 x 380
Functional Test	Factory beat rate target to meet maximum through put requirement for this entire process: Open ILM load plate --> Install or Remove an Intel LGA CPU --> Close ILM load plate (sec)	15	Not answered	ILM not suitable for PCBA test, the test is using automated fixture	12
	For Test CPU and Test Heatsink Installation at Functional Test: What is the time required for this entire operation?	When PCBA testing , we place fan on CPU w/o screw whole 4 corner	Not answered	OBE test typically complete within 10 sec	1 minute to complete

Key Message

- Most customers are performing bed of nails and functional tests
- Beat rate to install or remove an Intel LGA CPU from the socket ranges between 12-15 seconds

PCBA X-Ray



Process / Capability	ECS	LCFC	FXWH	LITEON	MSI (LENOVO)
Brand/Model	not answered	not answered	Brand: Dage, Model: XD7500, 2D	not answered	Brand: ELT/艾而特, Model: ST100A, 2D
Performed In-line / offline	Offline	Offline	Offline	Offline	not answered
Sampling rate	5 /day/line	5 / hrs	2 /6 hrs	3~5pcs/ hrs	not answered
Is the Detector - Camera based or Flat Panel?		not answered	Is based	Flat Panel	not answered
Detector Resolution (Mega Pixel)	1um	not answered	0.95um	Pixel:1004*1004 Campo di Visione (max): 102 x 102 mm	not answered
Built in filter to limit X ray Dosage & filter material/thickness (mm)?	1usv/h	not answered	None	None	not answered
What is the maximum angle (+/- degrees) for Oblique (2.5D) viewing?	70 angle, (2.5D) viewing	not answered	0-70°	140° (Right:70°, Left:70°)	not answered
What is the minimum distance of Xray tube Source to Object (BGA device)? (µm)	not answered	not answered	0.95	2000	not answered
Is the Source emitting X ray radiation applied from top to BGA device or is it from bottom below the PCB?	not answered	not answered	PCB bottom side	PCB top side	not answered
Is there a limitation specification for X-Ray (RADS/mGray) Dosage exposure?	< 5uSv/h	not answered	Yes	from bottom below the PCB	not answered
What is the cumulative specification for maximum exposure in RADS/mGray per scan as measured by a dosometer?	not answered	not answered	1uSv/ h	< 1 µSv / h	not answered
Are Global metal shields used between X-ray source and target SOC/Memory devices to limit dosage exposure?	not answered	not answered	Yes	Yes	not answered

Key Message

- Customers are performing PCBA x-ray inspections Offline

Rework



Process Parameter / Capability		ECS	LCFC	FXWH	LITEON
Machine / Profile	Peak Temp Range (°C)	High Temp process (235-245) Low Temp process (175-185)	230-250	235-245	235-245
	Time above Liquidus (sec)	60-80	60-100	60-90	60-90
	Soak Temp Range and Time (sec)	90-120	not answered	90-120	90-120
	Max BGA weight capability (grams)	45mmX24mm	30g	not answered	N/A
	Is rework done in air or N2? If N2, what is O2 PPM?	N2, 0.3-0.5PPM	Air	not answered	Air
Flux LTS	Apply flux on PCB pads?	Yes	Yes	Yes	No
	Any plans to use LTS (low temperature solder) paste (e.g. peak reflow temp <= 190 Deg C)?	Yes	No	No	No
Process	Print solder paste directly to Solder balls?	Yes	No	Yes	No
	Print solder paste directly to PCB pads?	Yes	Yes	Yes	Yes
	Rework Stencil Thickness used (mm)?	0.12	0.13	0.15, 0.18	0.10
	Use a pallet to support the board on the tool or other kind of fixtures?	Yes	Yes	Yes	Yes
	Nozzle size used is the exactly same size of the package to repair bigger than it? (mm)	2, 3	not answered	2, 3	not answered
	Do you perform a manual site dress process or auto scavenging?	Semi-auto	not answered	Yes	N/A
Rework Yield Target		N/A ?	not answered	99.9% (1K DPM)	98%

Key Message

- Rework stencil thicknesses used: 0.10, 0.12, 0.13, 0.15, 0.18
- One customer performing rework in N2, others in Air
- Pallets are used for board support during rework
- Customers print solder paste directly to the PCB pads for rework (a few can also print onto the solder balls)

Others

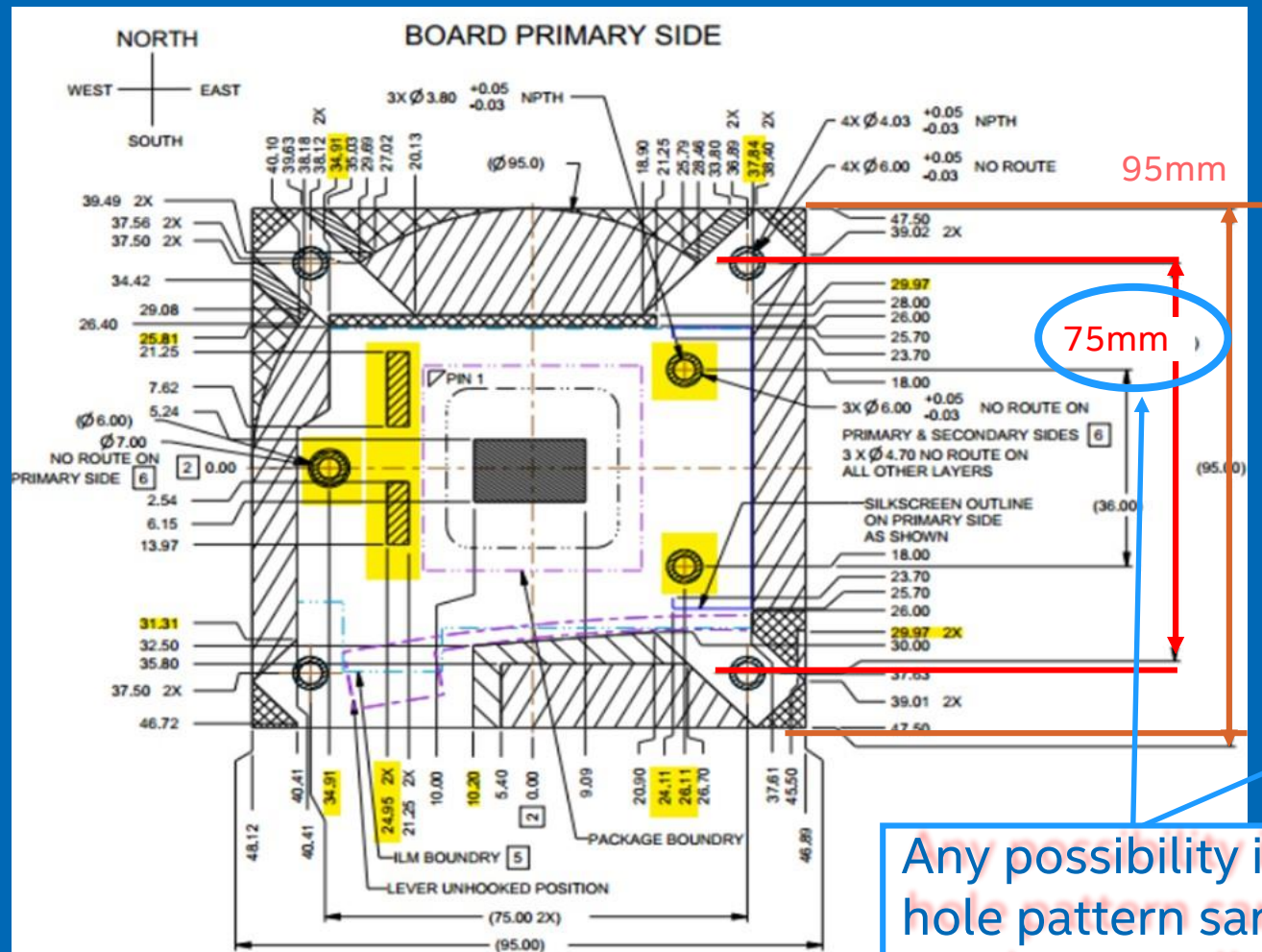


Process		ECS	LCFC	FXWH	LITEON
Misc	Do DDR or connectors have a different mfg. process than the Intel CPU?	not answered	not answered	Yes	No
Heatsink	How many times during the PCBA assembly process is the socket and package enabled and un-enabled (with a heatsink)?	not answered	not answered	Yes	1x
Shipping	How do you ship your boards out of the factory (with respect to the CPU and Heatsink installed)?	not answered	not answered	Without CPU	Without CPU
MAS / Video	Do you find the Desktop MAS helpful to understand the PCBA & System Assembly info and requirements?	not answered	not answered	Yes	Yes
	Do you think a System Assembly video (provided by Intel) is required or optional?	not answered	not answered	Yes	Yes

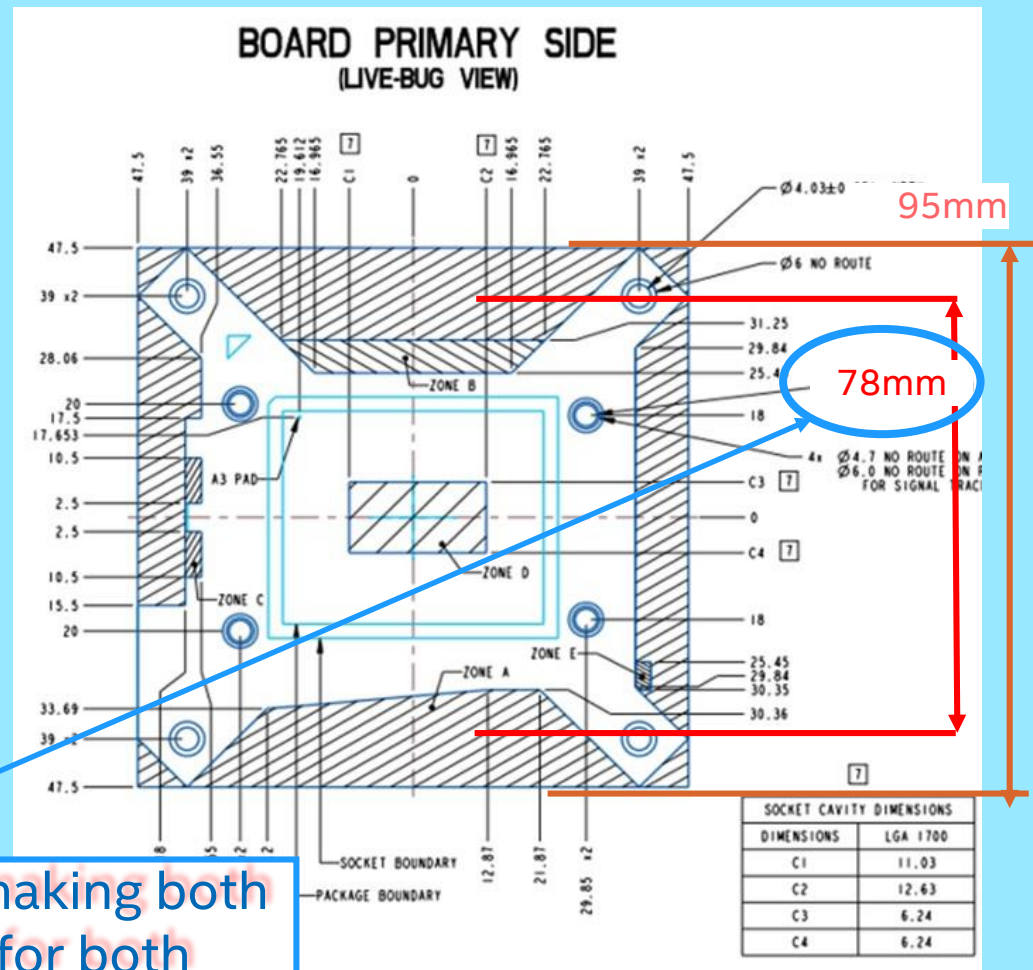
Key Message

- Customers find the Desktop MAS helpful and think a System Assembly video is required (customer prefer actual operator performing not animated)

Socket-H5 (CML-S/RKL-S)

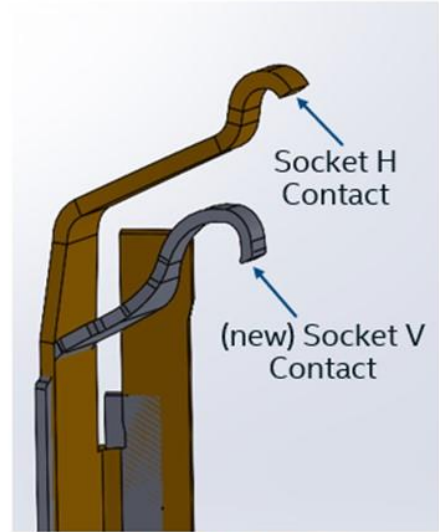


Socket-V (ADL-S)



Any possibility in making both hole pattern same for both products to allow same heatsink used ?

Socket H5 vs. Socket V0 Comparison

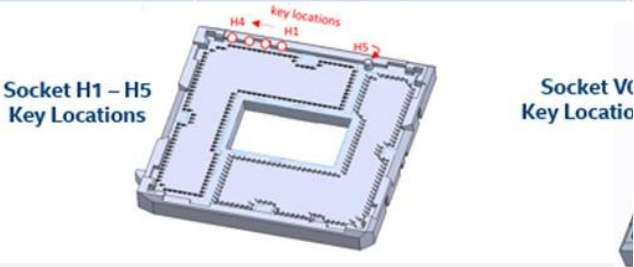


Socket Contact Tip Geometry

Note: The new Socket V contact tip shape is improved / more rounded than previous Socket H contact design and should be less susceptible to snags that cause bent pins.

Note: Values are preliminary and are subject to change. For latest values consult the respective component design guide or product specifications.

Attributes	Socket H5 (LGA1200)	Socket V0 (LGA1700)
LGA Processor Family	Comet Lake-S, Rocket Lake-S	Alder Lake-S and Future Processors
LGA Package X, Y Dimensions (mm)	37.5 x 37.5	37.5 x 45
LGA Package Thickness (mm)	Comet Lake-S 10+2: 4.454 ± 0.117 Rocket Lake-S 10+1: 4.282 ± 0.133	Alder Lake-S 881: 4.342 ± 0.116
Socket Pin Count	1200	1700
Socket X, Y, Z (mm) [Z is defined from skt seating plane to top of motherboard]	42.5 x 42.5 x 3.4	42.5 x 50 x 2.7
Socket Contact Nominal Pitch (mm)	0.914 x 0.914	0.8 x 0.8
Socket Cavity Size (mm)	11.9 x 19.6	13.5 x 24.7
Socket Keying to Processor	1 set (2 notches) of keying posts	2 set (4 notches) keying posts (see picture)
Socket Pin Distance to Side Wall (mm)	3.0	3.6
Socket Contact Tip	Same contact tip geometry for Skt H1-H5	Improved (see picture)
Socket Solderball Size (mm)	0.6	0.5 (FIT*/Deren*) or 0.45 (Lotes*)
Socket Solderball Alloy	Lead Free SAC305	Same as Socket H
Socket Contact Material / Plating	High Strength Copper Alloy / Min 1.27µm Ni all over, 0.381µm Au on Contact Tip	
Socket / ILM Suppliers	FIT*, Lotes*	
Socket Tray	18 sockets/tray (3x6)	

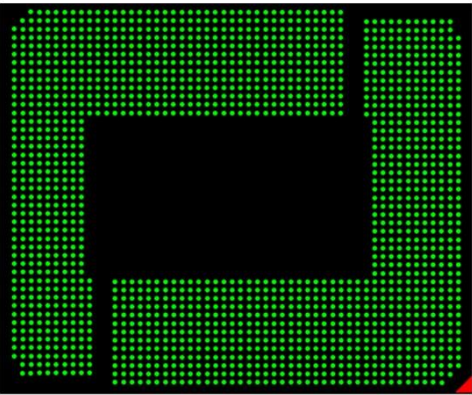


LOTES solder ball size is smaller than Deren & FIT, customer's concerns with SJQ with the same paste volume recommended in MAS across all socket supplier.

3.2.2 Solder Paste Volume and Stencil Optimization

Stencil Design Recommendation – Socket V0 (LGA1700-0) Center Volume Paste

Stencil Thickness: 127 µm (5 mils) option			
Stencil Design (Aperture)	Over-printing	Solder Paste Volume Target	Area Ratio
All Pads - Round 432 µm (17 mils)	Yes	0.0186 cu. mm (1135 cu. mils)	0.85



Alert:

- This aperture size shown here may not be optimized for all socket suppliers.
- Intel is currently investigating a larger aperture size stencil to mitigate any HnP or Opens risk, in the Air Reflow Environment

Note: This is only a PRELIMINARY recommendation (not final). It is subject to change as we continue to optimize paste volume for all of the different socket suppliers.

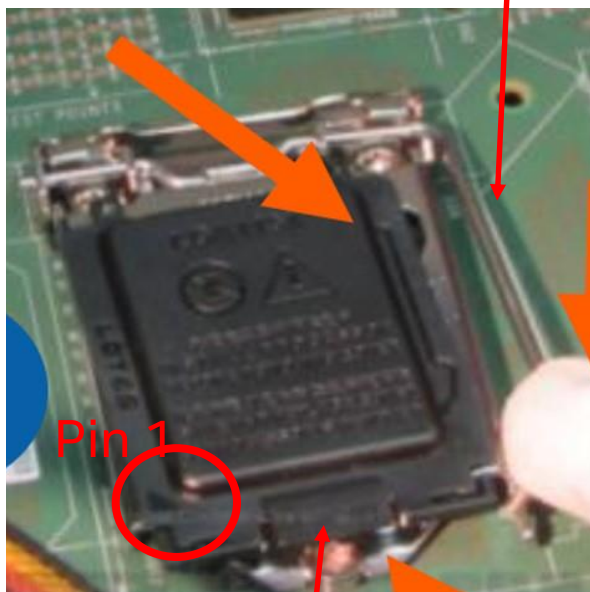
Notes: Reference Information Only.

- Design stencil opening to meet the theoretical volume requirements (above).
- A minimum of ± 203.2 µm (8 mils) air gap is needed, between adjacent stencil apertures, to prevent solder joint bridging. Stencil apertures can be rotated to meet this minimum requirement.
- Always consider the impact to via capping requirements when enlarging apertures beyond 1:1.
- Stencil design aligns with this component's PCB Land Pattern Design in Module 2.

Socket-H

Lever latch inward
(with right hand)

Consistent Pin 1
indication on tools and
PCB

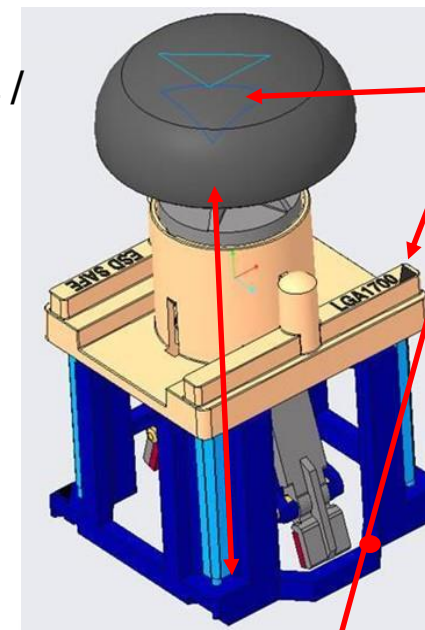


This site facing operator
(it will be inconvenient for same operator to process
skt-H and skt-V, and possibly error in execution, due
to the need to rotate the processor 180° orientation)

Socket-V

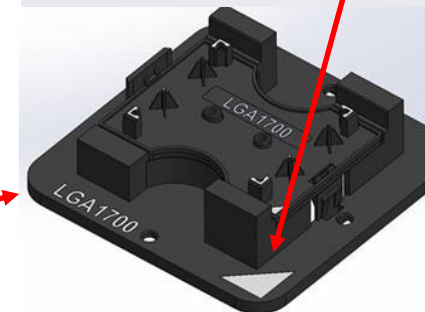
Lever latch
outward (with
left hand)

Pin 1, diagonally
opposite compare to
skt-H & also PnP tool /
stage



Which Pin 1
label is
correct ?

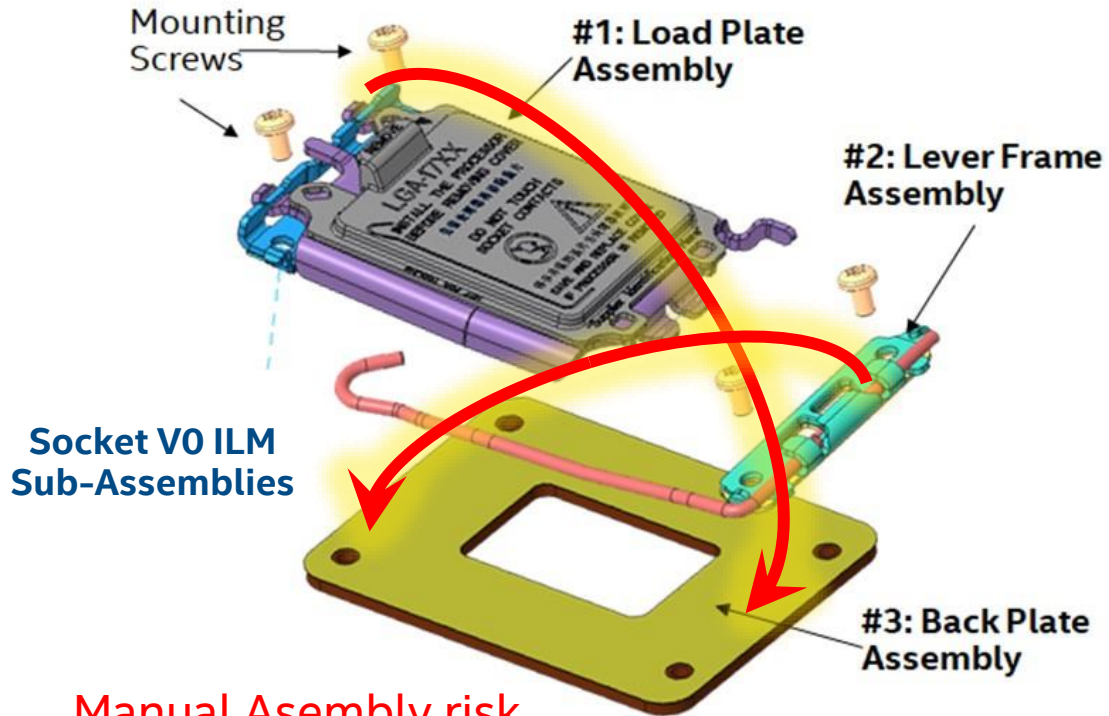
The Pin 1 mark
on tools are
inconsistent to
Pin 1 indicator
on PCB



Note:

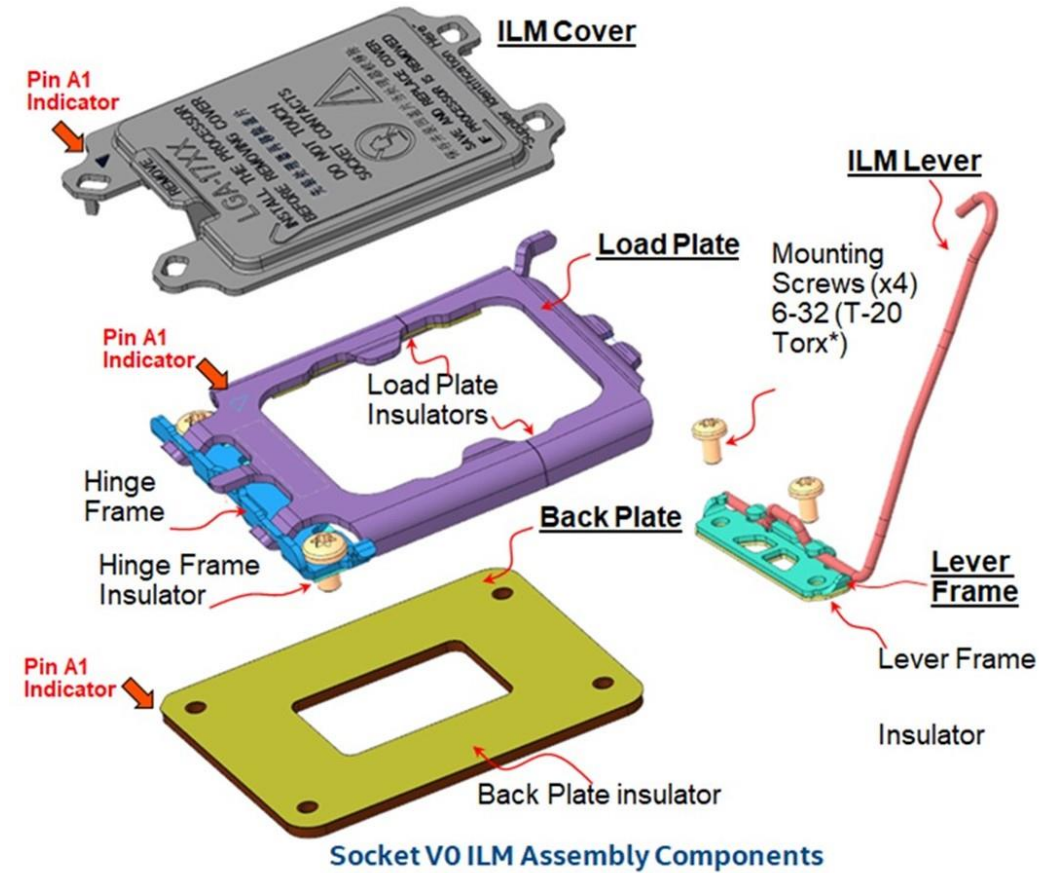
- 1) Operator needs to rotate processor 180° from PnP tool/stage in order to align to PCB Pin 1.
- 2) Any possibility to make changes for socket-V the ILM to allow customer process transparency and mitigate execution error for manual process ?eg. 180° marking orientation change on ILM cover marking

Skt-H, fool proof for wrong orientation



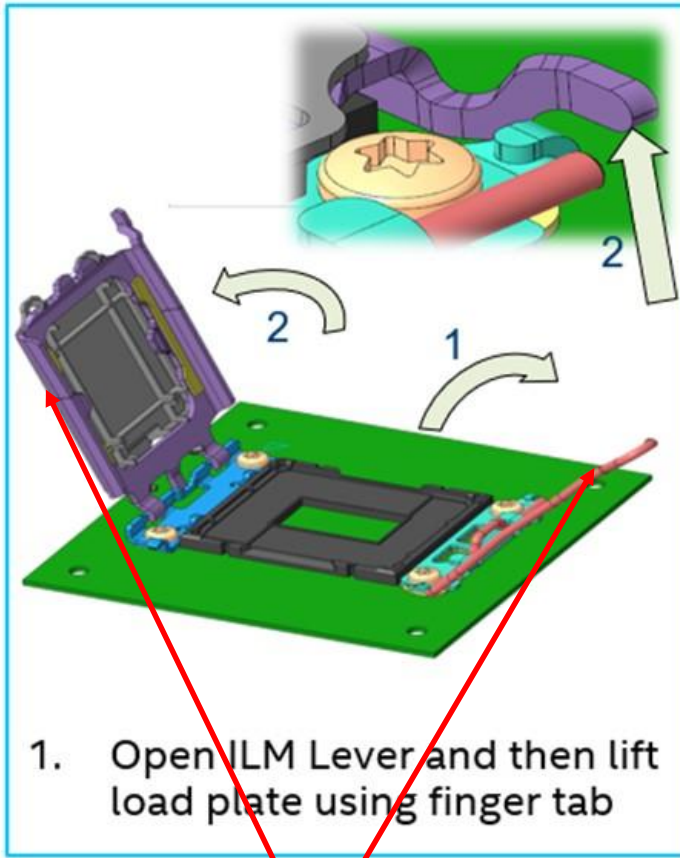
Manual Assembly risk

- 1) PCB Top Side> Potential/risk for operator swapped lever frame and load plate locations during assembly.
- 2) Backplate> Potentially 180° orientation assembled.

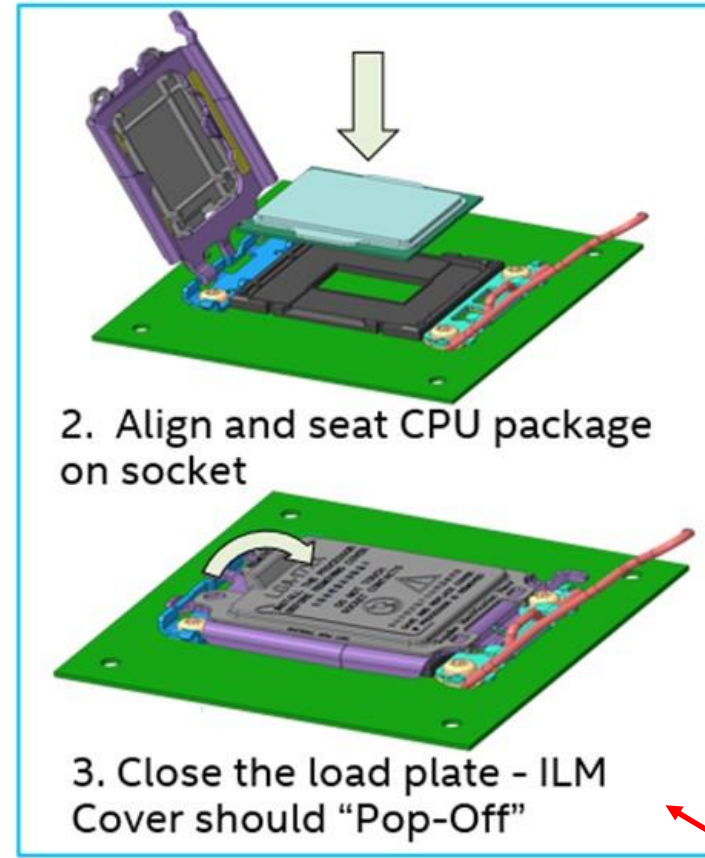


Piece Parts Interchangability

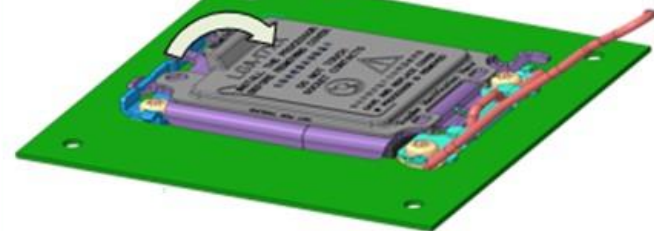
- 1) Need to ensure all 4 mounting screws are same part number# .
- 2) Allow mixed vendor for piece parts (including different socket vendors to ILM vendors).



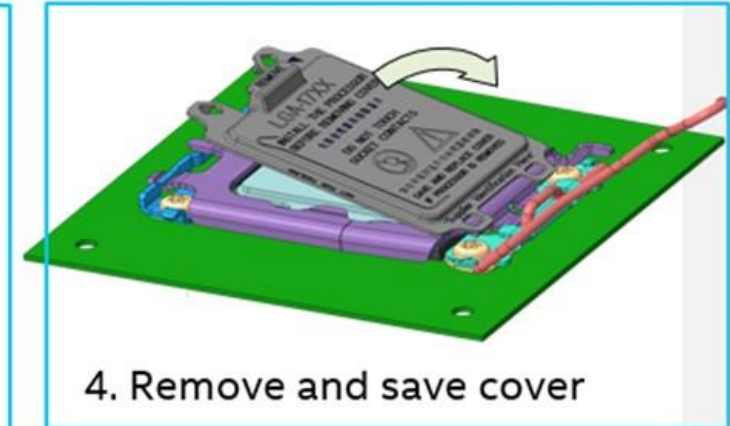
1. Open ILM Lever and then lift load plate using finger tab



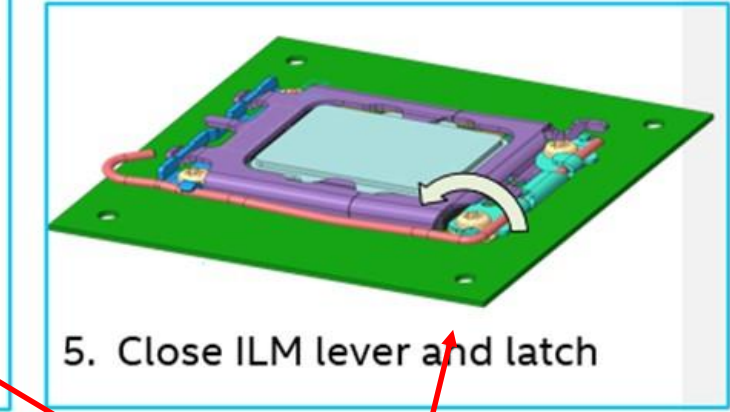
2. Align and seat CPU package on socket



3. Close the load plate - ILM Cover should "Pop-Off"



4. Remove and save cover



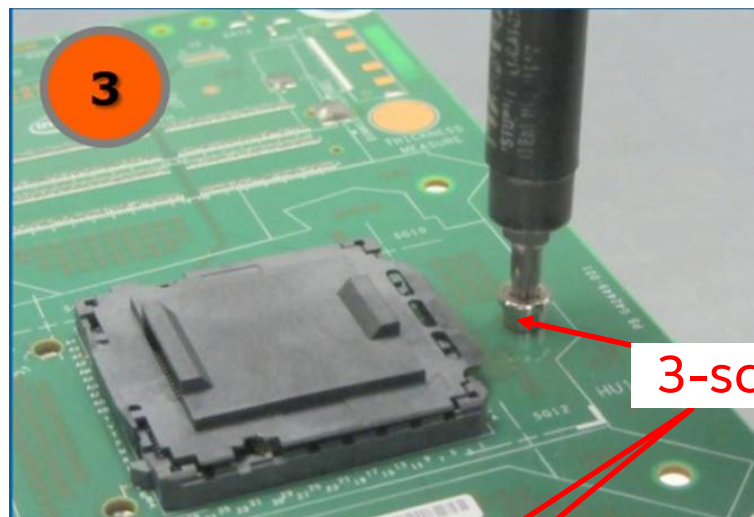
5. Close ILM lever and latch

Automation Assessment/concerns:

- 1) Will the top plate & lever arm will stay permanently (height & angle) when moving at conveyor belt to next process/operator to install processor ?
- 2) What is the height / angle of the top-plate and lever arm ?
(Machine PnP assessment)

Where is the locking position for load plate to push the ILM cover to pop-up ? Should the cover pop-up happen after lever is latch ?

Socket-H

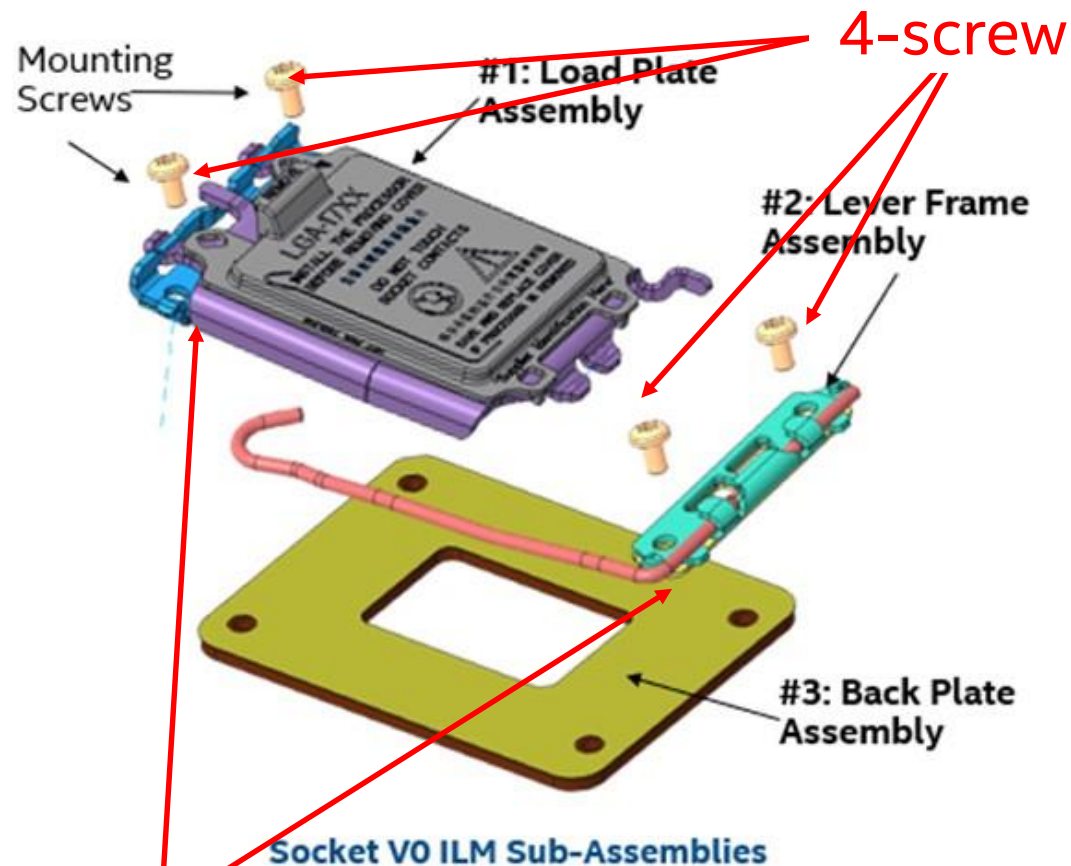


3-screw



1-pcs ILM top plate

Socket-V



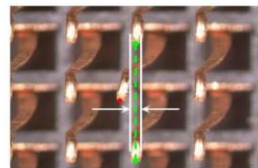
2-pcs ILM : top plate and lever frame

Customer needs to mix vendors for backplate, load plate, lever frame, screws & etc, is this allowable ?

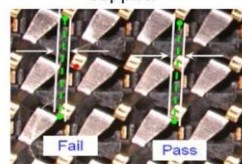
Socket-H MAS CDI# 561049

Contact Failure Type 3: X-Axis Shift

- Contact is shifted sideways (X axis direction)
- Reject contacts when the tip is shifted **more than 1 width** of the contact
- Reject **ALL** Type 3 failures



Geometry varies by supplier



Note: Red circle indicates contact out of alignment

>1 width of contact

CQR | Customer Quality & Regulatory

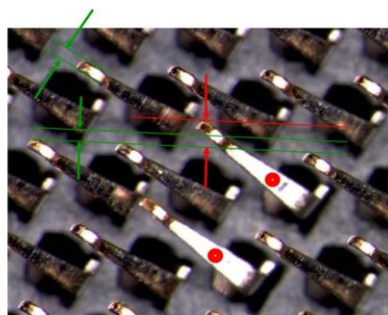
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Contact Failure Type 3 (continued): Y-Axis Shift

- Contact is shifted sideways (Y axis direction)
- Reject contacts when the tip is shifted **more than 1 width** of the contact
- Reject **ALL** Type 3 failures



Note: Red circles indicate contact out of alignment

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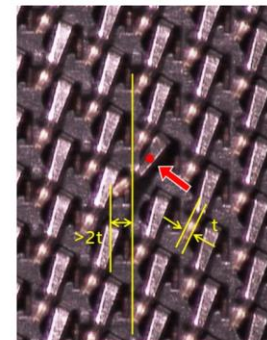
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Socket-V, Socket-P and Socket P4/5 MAS

6.7.6 Contact Failure Type 3: X-Axis Shift

- Contact is shifted sideways (X direction)
- Reject contacts when the tip is shifted **more than $\pm 0.125\text{mm}$** (approximately ± 1.5 width of the contact tip)
- Reject **ALL** Type 3 failures



Note: Red circle indicates contact out of alignment

$>\pm 0.125\text{mm}$ or ± 1.5 width of contact

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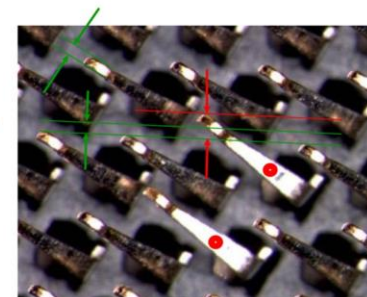
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6.7.6 Contact Failure Type 3: Y-Axis Shift

- Type 3 (continued)
 1. Contact is shifted sideways (Y axis direction)
 2. Reject contacts when the tip is shifted **more than $\pm 0.125\text{mm}$** (approximately ± 1.5 width of the contact tip)
 3. Reject **ALL** Type 3 failures



Note: Red circles indicate contact out of alignment

Tighter pitch but more relax criteria ?

- original >1 width vs current >1.5 width

Is the pin contact always contacted at the center of the LGA pad or near pad edge ? with the process variation and X-Y shift, will the pin be touching LGA Au pad or outside the Au pad (Solder resist area) or adjacent Au pad ? -- Need Intel design to assess

